



Practice Activity

Objectives

In this activity you will design and write code for an application for a training company. The app will be used by the staff and customers to perform basic admin tasks such as:

1. **Register new students**
Record basic details such as name, age, company, level (3 to 6) and stream (development, testing, cyber security, networking). In future, we might add new streams.
2. **Register new trainers and keep a list of their skills area**
Record basic details such as name, preferred cities to teach a class and technical areas such as programming, networking, cyber security, testing and digital marketing. In future we might add new skills areas.
3. **Register a course**
Save details such as name, duration and skills required to teach the course.
4. QA has a calendar of course runs which indicate when a course will run. This is created off-line by a group of skilled admin personnel.
5. A registered student should be able to:
 - a. View a list of courses by type
 - b. Choose a particular course and then view when it will be running
 - c. The student can then register on a chosen course run
 - d. Detail of this transaction is then recorded in a database
 - e. Payment must be made for each booked course run
 - f. Of course, the student is free to cancel a booking and get the money back

Your task

1. It is always good to start with a simple use case diagram in order to identify all the tasks (use cases) and actors for your system.
2. Identify the microservices which are required to implement the app.
If you feel a microservice is becoming a monolith then break it up into several microservices.
3. Group related microservices into domains.
4. Identify any dependency between services (services using other services).



5. Identify what type of data store is required for each service based on the speed of access, frequency of transactions and any other relevant reason. Is there a need for a single central database?
6. Which of the services require tighter security and how would you provide security?
7. Decide which type of database you will need and state why you have chosen the type.
There are many database types such as MySQL, MS-SQL, NoSQL, or even file-based data store, etc. Each of these databases has pros and cons and there is no single database type that satisfies every requirement.
8. Finally, write code for at least two of the services.
9. Create unit and integration tests to make sure your code works.
How would you test modules who have dependency on services that grant access to a live database?

You may choose any language you feel comfortable with, but we suggest using Java, C# or Python.

10. (Optional) Create a simple UI to run these outside of a test application.

**** End ****

Submission of your work

Before starting to work on this activity, submit a simple plan explaining what you are going to do. This will ensure that the piece of work selected meets all the criteria required at this stage.

Be sure to submit your plan within three days. Here are some guidelines:

The submission should describe:

- The project and context description
- The support available to you (e.g. from your line manager or wider team)
- The planned outputs

Submission Format

To submit your reflection, please follow these steps.



Instructions & Format

Your submission should be in written format.

There is not a set template for this task, so you can structure your submission in any way you want. However, we would suggest something that includes the following:

Introduction (one paragraph):

- Explain the project's context (why is it important?)
- Write in the first person (use 'I')
- Explain your personal input

Conclusion (one paragraph):

Use reflective questions such as:

- What have I learnt?
- What went well?
- What did not go well?
- What will I do the same next time?
- What will I do differently next time?

Evidence:

You should include supporting evidence / material such as:

- Diagrams
- Photographic evidence (e.g. hand-drawn sketches)
- Prototypes
- Screenshots (e.g. designs, code, or the working app)
- Video evidence (e.g. screen recording of working app, if needed)
- Written documentation

You can also include appendix items or screen prints using word processing software.

How to submit your work?

- Go to Bud and locate your activity window
- Click on the 'Submissions and Messages' button
- Click on the 'Submit Work/Message'



- Write a message with your answers (you may submit a written document too. However, we would like to encourage you to keep this submission as simple as possible)
- Click on the 'Submit' button

We will be in touch with you shortly. In the meantime, if you need any support, please reach out to us at any time.